

Assignment 2 (Conditions – Loop)

Submission guideline:

1. SOLVE ALL 35 Problems
2. You have to write each program in separate c file. Suppose your student ID – 0112019344 and your section is "U"
Then the name of your files will be –
0112019344_32.c // for problem 32
0112019344_33.c // for problem 33
0112019344_34.c // for problem 34
0112019344_35.c // for problem 35
0112019344_36.c // for problem 36
0112019344_37.c // for problem 37
3. Then put all the c files(**only .c files not .exe or .o**) in one folder and **rename the folder with your "student ID_Assignment02_Section_ICS_Fall23"**
4. Zip the folder and finally submit the **0112019344_Assignment02_U_ICS_Fall23.zip** file.
5. Submission deadline : **12 December 2023 12:00 AM**
6. Please do not copy codes from others or directly from the internet. Each of the assignments will be evaluated with a viva. You must be able to explain your code. Also, we will run a copy checker on the submissions. Any plagiarism will be severely penalized.

Condition Related Problems

(Total 15 questions)

SL	Problem statement	Difficulty levels										
32.	Program that will decide whether a number is positive or not.	*										
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>100</td><td>Positive</td></tr><tr><td>-11.11</td><td>Negative</td></tr><tr><td>0</td><td>Positive</td></tr></table>		Sample input	Sample output	100	Positive	-11.11	Negative	0	Positive		
	Sample input		Sample output									
	100		Positive									
	-11.11		Negative									
	0		Positive									
33.	Program that will decide whether a number is even or odd.	*										
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>50</td><td>Even</td></tr><tr><td>-77</td><td>Odd</td></tr><tr><td>0</td><td>Even</td></tr></table>		Sample input	Sample output	50	Even	-77	Odd	0	Even		
	Sample input		Sample output									
	50		Even									
	-77		Odd									
	0		Even									
34.	Program that will take an integer of length one from the terminal and then display the digit in English.	*										
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>9</td><td>nine</td></tr><tr><td>0</td><td>zero</td></tr></table>		Sample input	Sample output	9	nine	0	zero				
	Sample input		Sample output									
	9		nine									
	0		zero									
35.	Program that will check whether a triangle is valid or not, when the three angles (angle value should be such that, $0 < \text{value} < 180$) of the triangle are entered through the keyboard. [Hint: A triangle is valid if the sum of all the three angles is equal to 180 degrees.]	*										
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>90 45 45</td><td>Yes</td></tr><tr><td>30 110 40</td><td>Yes</td></tr><tr><td>160 20 30</td><td>No</td></tr><tr><td>0 180 0</td><td>No</td></tr></table>		Sample input	Sample output	90 45 45	Yes	30 110 40	Yes	160 20 30	No	0 180 0	No
	Sample input		Sample output									
	90 45 45		Yes									
	30 110 40		Yes									
	160 20 30		No									
	0 180 0		No									

36.	Program that will read from the console a random positive nonzero number and determine if it is a power of 2. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>1</td><td>Yes</td></tr><tr><td>512</td><td>Yes</td></tr><tr><td>1022</td><td>No</td></tr></table>	Sample input	Sample output	1	Yes	512	Yes	1022	No	**				
Sample input	Sample output													
1	Yes													
512	Yes													
1022	No													
37.	Program that will read from the console a random number and check if it is a nonzero positive number. If the check is yes, it will determine if the number is a power of 2. If the check fails the program will check for two more cases. If the number is zero, the program will print “Zero is not a valid input”. Else it will print “Negative input is not valid”. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>0</td><td>Zero is not a valid input</td></tr><tr><td>1</td><td>Yes</td></tr><tr><td>512</td><td>Yes</td></tr><tr><td>1022</td><td>No</td></tr><tr><td>-512</td><td>Negative input is not valid</td></tr></table>	Sample input	Sample output	0	Zero is not a valid input	1	Yes	512	Yes	1022	No	-512	Negative input is not valid	***
Sample input	Sample output													
0	Zero is not a valid input													
1	Yes													
512	Yes													
1022	No													
-512	Negative input is not valid													
38.	Program that will take two numbers X & Y as inputs and decide whether X is greater than/less than/equal to Y. <table><tr><th>Sample input (X,Y)</th><th>Sample output</th></tr><tr><td>5 -10</td><td>5 is greater than -10</td></tr><tr><td>5 10</td><td>5 is less than 10</td></tr><tr><td>5 5</td><td>5 is equal to 5</td></tr></table>	Sample input (X,Y)	Sample output	5 -10	5 is greater than -10	5 10	5 is less than 10	5 5	5 is equal to 5	*				
Sample input (X,Y)	Sample output													
5 -10	5 is greater than -10													
5 10	5 is less than 10													
5 5	5 is equal to 5													
39.	Program that will decide whether a year is leap year or not. Yes, if (Year % 4 == 0 && year % 100 != 0) (Year % 400 ==0) <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>2000</td><td>Yes</td></tr><tr><td>2004</td><td>Yes</td></tr><tr><td>2014</td><td>No</td></tr></table>	Sample input	Sample output	2000	Yes	2004	Yes	2014	No	*				
Sample input	Sample output													
2000	Yes													
2004	Yes													
2014	No													

40.	Program that will categorize a single character that is entered at the terminal, whether it is an alphabet, a digit or a special character. (Restriction: Without math.h) <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>z</td><td>Alphabet</td></tr><tr><td>A</td><td>Alphabet</td></tr><tr><td>8</td><td>Digit</td></tr><tr><td>*</td><td>Special</td></tr></table>	Sample input	Sample output	z	Alphabet	A	Alphabet	8	Digit	*	Special	*																																
Sample input	Sample output																																											
z	Alphabet																																											
A	Alphabet																																											
8	Digit																																											
*	Special																																											
41.	Program that will evaluate simple expressions of the form- <number1> <operator> <number2> ; where operators are (+, - , *, /) And if the operator is “/”, then check if <number2> nonzero or not. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>100 * 55.5</td><td>Multiplication: 5550</td></tr><tr><td>100 / -5.5</td><td>Division: -18.181818</td></tr><tr><td>100 / 0</td><td>Division: Zero as divisor is not valid!</td></tr></table>	Sample input	Sample output	100 * 55.5	Multiplication: 5550	100 / -5.5	Division: -18.181818	100 / 0	Division: Zero as divisor is not valid!	**																																		
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100 * 55.5	Multiplication: 5550																																											
100 / -5.5	Division: -18.181818																																											
100 / 0	Division: Zero as divisor is not valid!																																											
42.	Program that will take the final score of a student in a particular subject as input and find his/her grade. <table><tr><th>Marks</th><th>Letter Grade</th><th>Marks</th><th>Letter Grade</th><th>Marks</th><th>Letter Grade</th></tr><tr><td>90-100</td><td>A</td><td>70-73</td><td>C+</td><td>Less than 55</td><td>F</td></tr><tr><td>86-89</td><td>A-</td><td>66-69</td><td>C</td><td></td><td></td></tr><tr><td>82-85</td><td>B+</td><td>62-65</td><td>C-</td><td></td><td></td></tr><tr><td>78-81</td><td>B</td><td>58-61</td><td>D+</td><td></td><td></td></tr><tr><td>74-77</td><td>B-</td><td>55-57</td><td>D</td><td></td><td></td></tr></table> <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>91.5</td><td>Grade: A</td></tr><tr><td>50</td><td>Grade: F</td></tr></table>	Marks	Letter Grade	Marks	Letter Grade	Marks	Letter Grade	90-100	A	70-73	C+	Less than 55	F	86-89	A-	66-69	C			82-85	B+	62-65	C-			78-81	B	58-61	D+			74-77	B-	55-57	D			Sample input	Sample output	91.5	Grade: A	50	Grade: F	*
Marks	Letter Grade	Marks	Letter Grade	Marks	Letter Grade																																							
90-100	A	70-73	C+	Less than 55	F																																							
86-89	A-	66-69	C																																									
82-85	B+	62-65	C-																																									
78-81	B	58-61	D+																																									
74-77	B-	55-57	D																																									
Sample input	Sample output																																											
91.5	Grade: A																																											
50	Grade: F																																											

43.	Program that will construct a menu for performing arithmetic operations. The user will give two real numbers (a, b) on which the arithmetic operations will be performed and an integer number ($1 \leq \text{Choice} \leq 4$) as a choice. Choice-1, 2, 3, 4 are for performing addition, subtraction, multiplication, division (quotient) respectively. <table><tr><th>Sample input (a, b, Choice)</th><th>Sample output</th></tr><tr><td>5 10 3</td><td>Multiplication: 50</td></tr><tr><td>-5 10.5 4</td><td>Quotient: 0</td></tr></table>	Sample input (a, b, Choice)	Sample output	5 10 3	Multiplication: 50	-5 10.5 4	Quotient: 0	*		
Sample input (a, b, Choice)	Sample output									
5 10 3	Multiplication: 50									
-5 10.5 4	Quotient: 0									
44.	Program that will construct a menu for performing arithmetic operations. The user will give two real numbers (a, b) on which the arithmetic operations will be performed and an integer number ($1 \leq \text{Choice} \leq 4$) as a choice. Choice : 1, 2, 3, 4 are for performing addition, subtraction, multiplication, division respectively. If Choice: 4 is selected, again the program will ask for another choice ($1 \leq \text{Case} \leq 2$), where Case: 1, 2 evaluate quotient and remainder respectively. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>5 10 3</td><td>Multiplication: 50</td></tr><tr><td>-5 10.5 4 1</td><td>Quotient: 0</td></tr><tr><td>-5 10.5 4 2</td><td>Remainder: -48</td></tr></table> 1. Addition 2. Subtraction 3. Multiplication 4. Division 1. Quotient 2. Remainder	Sample input	Sample output	5 10 3	Multiplication: 50	-5 10.5 4 1	Quotient: 0	-5 10.5 4 2	Remainder: -48	**
Sample input	Sample output									
5 10 3	Multiplication: 50									
-5 10.5 4 1	Quotient: 0									
-5 10.5 4 2	Remainder: -48									

45.	Program that will construct a menu for performing arithmetic operations. The user will give two real numbers (a, b) on which the arithmetic operations will be performed and an integer number ($1 \leq \text{Choice} \leq 4$) as a choice. Choice-1, 2, 3, 4 are for performing addition, subtraction, multiplication, division respectively. If Choice-4 is selected, the program will check if b is nonzero. If the check is true, the program will ask for another choice ($1 \leq \text{Case} \leq 2$), where Case-1, 2 evaluate quotient and remainder respectively. If the check is false, it will print an error message "Error: Divisor is zero" and halt. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>5 10 3</td><td>Multiplication: 50</td></tr><tr><td>-5 10.5 4 2</td><td>Reminder: -48</td></tr><tr><td>-5 0 4</td><td>Error: Divisor is zero</td></tr></table>	Sample input	Sample output	5 10 3	Multiplication: 50	-5 10.5 4 2	Reminder: -48	-5 0 4	Error: Divisor is zero	***
Sample input	Sample output									
5 10 3	Multiplication: 50									
-5 10.5 4 2	Reminder: -48									
-5 0 4	Error: Divisor is zero									
46.	Program for "Guessing Game": Player-1 picks a number X and Player-2 has to guess that number within N = 3 tries. For each wrong guess by Player-2, the program prints "Wrong, N-1 Chance(s) Left!" If Player-2 successfully guesses the number, the program prints "Right, Player-2 wins!" and <u>stops allowing further tries (if any left)</u>. Otherwise after the completion of N = 3 wrong tries, the program prints "Player-1 wins!" and halts. [Restriction: Without using loop/break/continue Hint: Use flag] <table><tr><th>Sample input (X, n1, n2, n3)</th><th>Sample output</th></tr><tr><td>5 12 8 5</td><td>Wrong, 2 Chance(s) Left! Wrong, 1 Chance(s) Left! Right, Player-2 wins!</td></tr><tr><td>100 50 100</td><td>Wrong, 2 Chance(s) Left! Right, Player-2 wins!</td></tr><tr><td>20 12 8 5</td><td>Wrong, 2 Chance(s) Left! Wrong, 1 Chance(s) Left! Wrong, 0 Chance(s) Left! Player-1 wins!</td></tr></table>	Sample input (X, n1, n2, n3)	Sample output	5 12 8 5	Wrong, 2 Chance(s) Left! Wrong, 1 Chance(s) Left! Right, Player-2 wins!	100 50 100	Wrong, 2 Chance(s) Left! Right, Player-2 wins!	20 12 8 5	Wrong, 2 Chance(s) Left! Wrong, 1 Chance(s) Left! Wrong, 0 Chance(s) Left! Player-1 wins!	***
Sample input (X, n1, n2, n3)	Sample output									
5 12 8 5	Wrong, 2 Chance(s) Left! Wrong, 1 Chance(s) Left! Right, Player-2 wins!									
100 50 100	Wrong, 2 Chance(s) Left! Right, Player-2 wins!									
20 12 8 5	Wrong, 2 Chance(s) Left! Wrong, 1 Chance(s) Left! Wrong, 0 Chance(s) Left! Player-1 wins!									

Loop related problems (total 20 questions)

SL	Problem statement	Difficulty levels														
47.	Write a program (WAP) that will print following series upto N th terms. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,	*														
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>2</td><td>1, 2</td></tr><tr><td>5</td><td>1, 2, 3, 4, 5</td></tr><tr><td>11</td><td>1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11</td></tr></table>		Sample input	Sample output	2	1, 2	5	1, 2, 3, 4, 5	11	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11						
	Sample input		Sample output													
	2		1, 2													
	5		1, 2, 3, 4, 5													
	11		1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11													
48.	Write a program (WAP) that will print following series upto N th terms. 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31	*														
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>2</td><td>1, 3</td></tr><tr><td>5</td><td>1, 3, 5, 7, 9</td></tr><tr><td>11</td><td>1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21</td></tr></table>		Sample input	Sample output	2	1, 3	5	1, 3, 5, 7, 9	11	1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21						
	Sample input		Sample output													
	2		1, 3													
	5		1, 3, 5, 7, 9													
	11		1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21													
49.	Write a program (WAP) that will print following series upto N th terms. 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1,	**														
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1, 0</td></tr><tr><td>3</td><td>1, 0, 1</td></tr><tr><td>4</td><td>1, 0, 1, 0</td></tr><tr><td>7</td><td>1, 0, 1, 0, 1, 0, 1</td></tr><tr><td>13</td><td>1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1</td></tr></table>		Sample input	Sample output	1	1	2	1, 0	3	1, 0, 1	4	1, 0, 1, 0	7	1, 0, 1, 0, 1, 0, 1	13	1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1
	Sample input		Sample output													
	1		1													
	2		1, 0													
	3		1, 0, 1													
	4		1, 0, 1, 0													
	7		1, 0, 1, 0, 1, 0, 1													
	13		1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1													
50.	Write a program (WAP) that will take N numbers as inputs and compute their average. (Restriction: Without using any array)	*														
	<table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>3 10 20 30.5</td><td>AVG of 3 inputs: 20.166667</td></tr><tr><td>2 22.4 11.1</td><td>AVG of 2 inputs: 16.750000</td></tr><tr><td></td><td></td></tr></table>		Sample input	Sample output	3 10 20 30.5	AVG of 3 inputs: 20.166667	2 22.4 11.1	AVG of 2 inputs: 16.750000								
	Sample input		Sample output													
	3 10 20 30.5		AVG of 3 inputs: 20.166667													
	2 22.4 11.1		AVG of 2 inputs: 16.750000													

51.	Write a program (WAP) that will take two numbers X and Y as inputs. Then it will print the square of X and increment (if X<Y) or decrement (if X>Y) X by 1, until X reaches Y. If and when X is equal to Y, the program prints “Reached!” <table><tr><th><i>Sample input(X,Y)</i></th><th><i>Sample output</i></th></tr><tr><td>10 5</td><td>100, 81, 64, 49, 36, Reached!</td></tr><tr><td>5 10</td><td>25, 36, 49, 64, 81, Reached!</td></tr><tr><td>10 10</td><td>Reached!</td></tr></table>	<i>Sample input(X,Y)</i>	<i>Sample output</i>	10 5	100, 81, 64, 49, 36, Reached!	5 10	25, 36, 49, 64, 81, Reached!	10 10	Reached!	*
<i>Sample input(X,Y)</i>	<i>Sample output</i>									
10 5	100, 81, 64, 49, 36, Reached!									
5 10	25, 36, 49, 64, 81, Reached!									
10 10	Reached!									
52.	Write a program (WAP) for the described scenario: Player-1 picks a number X and Player-2 has to guess that number within N tries. For each wrong guess by Player-2, the program prints “Wrong, N-1 Choice(s) Left!” If Player-2 at any time successfully guesses the number, the program prints “Right, Player-2 wins!” and <u>terminates right away</u>. Otherwise after the completion of N wrong tries, the program prints “Player-1 wins!” and halts. (Hint: Use break/continue) <table><tr><th><i>Sample input (X,N,n1, n2,...,nN)</i></th><th><i>Sample output</i></th></tr><tr><td>5 3 12 8 5</td><td>Wrong, 2 Choice(s) Left! Wrong, 1 Choice(s) Left! Right, Player-2 wins!</td></tr><tr><td>100 5 50 100</td><td>Wrong, 4 Choice(s) Left! Right, Player-2 wins!</td></tr><tr><td>20 3 12 8 5</td><td>Wrong, 2 Choice(s) Left! Wrong, 1 Choice(s) Left! Wrong, 0 Choice(s) Left! Player-1 wins!</td></tr></table>	<i>Sample input (X,N,n1, n2,...,nN)</i>	<i>Sample output</i>	5 3 12 8 5	Wrong, 2 Choice(s) Left! Wrong, 1 Choice(s) Left! Right, Player-2 wins!	100 5 50 100	Wrong, 4 Choice(s) Left! Right, Player-2 wins!	20 3 12 8 5	Wrong, 2 Choice(s) Left! Wrong, 1 Choice(s) Left! Wrong, 0 Choice(s) Left! Player-1 wins!	**
<i>Sample input (X,N,n1, n2,...,nN)</i>	<i>Sample output</i>									
5 3 12 8 5	Wrong, 2 Choice(s) Left! Wrong, 1 Choice(s) Left! Right, Player-2 wins!									
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20 3 12 8 5	Wrong, 2 Choice(s) Left! Wrong, 1 Choice(s) Left! Wrong, 0 Choice(s) Left! Player-1 wins!									
53.	Write a program (WAP) that will run and show keyboard inputs until the user types an 'A' at the keyboard. <table><tr><th><i>Sample input</i></th><th><i>Sample output</i></th></tr><tr><td>X 1 a A</td><td>Input 1: X Input 2: 1 Input 3: a</td></tr></table>	<i>Sample input</i>	<i>Sample output</i>	X 1 a A	Input 1: X Input 2: 1 Input 3: a	*				
<i>Sample input</i>	<i>Sample output</i>									
X 1 a A	Input 1: X Input 2: 1 Input 3: a									

54.	Write a program (WAP) that will reverse the digits of an input integer. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>13579</td><td>97531</td></tr><tr><td>4321</td><td>1234</td></tr></table>	Sample input	Sample output	13579	97531	4321	1234	**																																																
Sample input	Sample output																																																							
13579	97531																																																							
4321	1234																																																							
55.	Write a program (WAP) that will find the grade of N students. For each student, it will take the marks of his/her the attendance (on 5 marks), assignment (on 10 marks), class test (on 15 marks), midterm (on 50 marks), term final (on 100 marks). Then based on the tables shown below, the program will output his grade. <table><tr><td>Attendance (A)</td><td>5%</td></tr><tr><td>Assignments (HW)</td><td>10%</td></tr><tr><td>Class Tests (CT)</td><td>15%</td></tr><tr><td>Midterm (MT)</td><td>30%</td></tr><tr><td>Final (TF)</td><td>40%</td></tr></table> <table><tr><td>Marks</td><td>Letter Grade</td><td>Marks</td><td>Letter Grade</td><td>Marks</td><td>Letter Grade</td></tr><tr><td>90-100</td><td>A</td><td>70-73</td><td>C+</td><td>Less than 55</td><td>F</td></tr><tr><td>86-89</td><td>A-</td><td>66-69</td><td>C</td><td></td><td></td></tr><tr><td>82-85</td><td>B+</td><td>62-65</td><td>C-</td><td></td><td></td></tr><tr><td>78-81</td><td>B</td><td>58-61</td><td>D+</td><td></td><td></td></tr><tr><td>74-77</td><td>B-</td><td>55-57</td><td>D</td><td></td><td></td></tr></table> <table><tr><th>Sample input (A,HW,CT,MT,TF)</th><th>Sample output</th></tr><tr><td>2</td><td>Student 1 : A</td></tr><tr><td>5 10 15 44.5 92.5</td><td>Student 2 : F</td></tr><tr><td>0 7.5 5 20 55.5</td><td></td></tr></table>	Attendance (A)	5%	Assignments (HW)	10%	Class Tests (CT)	15%	Midterm (MT)	30%	Final (TF)	40%	Marks	Letter Grade	Marks	Letter Grade	Marks	Letter Grade	90-100	A	70-73	C+	Less than 55	F	86-89	A-	66-69	C			82-85	B+	62-65	C-			78-81	B	58-61	D+			74-77	B-	55-57	D			Sample input (A,HW,CT,MT,TF)	Sample output	2	Student 1 : A	5 10 15 44.5 92.5	Student 2 : F	0 7.5 5 20 55.5		*
Attendance (A)	5%																																																							
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78-81	B	58-61	D+																																																					
74-77	B-	55-57	D																																																					
Sample input (A,HW,CT,MT,TF)	Sample output																																																							
2	Student 1 : A																																																							
5 10 15 44.5 92.5	Student 2 : F																																																							
0 7.5 5 20 55.5																																																								
56.	Write a program (WAP) that will give the sum of first Nth terms for the following series. 1, -2, 3, -4, 5, -6, 7, -8, 9, -10, 11, -12, 13, -14, <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>2</td><td>Result: -1</td></tr><tr><td>3</td><td>Result: 2</td></tr><tr><td>4</td><td>Result: -2</td></tr></table>	Sample input	Sample output	2	Result: -1	3	Result: 2	4	Result: -2	**																																														
Sample input	Sample output																																																							
2	Result: -1																																																							
3	Result: 2																																																							
4	Result: -2																																																							

57.	Write a program (WAP) that will calculate the result for the first Nth terms of the following series. [In that series sum, dot sign (.) means multiplication] $1^2.2 + 2^2.3 + 3^2.4 + 4^2.5 + \dots$ <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>2</td><td>Result: 14</td></tr><tr><td>3</td><td>Result: 50</td></tr><tr><td>4</td><td>Result: 130</td></tr><tr><td>7</td><td>Result: 924</td></tr></table>	Sample input	Sample output	2	Result: 14	3	Result: 50	4	Result: 130	7	Result: 924	**
Sample input	Sample output											
2	Result: 14											
3	Result: 50											
4	Result: 130											
7	Result: 924											
58.	Write a program (WAP) that will print Fibonacci series upto Nth terms. 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1, 1</td></tr><tr><td>4</td><td>1, 1, 2, 3</td></tr><tr><td>7</td><td>1, 1, 2, 3, 5, 8, 13</td></tr></table>	Sample input	Sample output	1	1	2	1, 1	4	1, 1, 2, 3	7	1, 1, 2, 3, 5, 8, 13	**
Sample input	Sample output											
1	1											
2	1, 1											
4	1, 1, 2, 3											
7	1, 1, 2, 3, 5, 8, 13											
59.	Write a program (WAP) that will print the factorial (N!) of a given number N. Please see the sample input output. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>1</td><td>1! = 1 = 1</td></tr><tr><td>2</td><td>2! = 2 X 1 = 2</td></tr><tr><td>3</td><td>3! = 3 X 2 X 1 = 6</td></tr><tr><td>4</td><td>4! = 4 X 3 X 2 X 1 = 24</td></tr></table>	Sample input	Sample output	1	1! = 1 = 1	2	2! = 2 X 1 = 2	3	3! = 3 X 2 X 1 = 6	4	4! = 4 X 3 X 2 X 1 = 24	**
Sample input	Sample output											
1	1! = 1 = 1											
2	2! = 2 X 1 = 2											
3	3! = 3 X 2 X 1 = 6											
4	4! = 4 X 3 X 2 X 1 = 24											
60.	Write a program (WAP) that will find ⁿC_r, where n >= r; n and r are integers. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>5 2</td><td>10</td></tr><tr><td>10 3</td><td>120</td></tr><tr><td>7 7</td><td>1</td></tr><tr><td>6 1</td><td>6</td></tr></table>	Sample input	Sample output	5 2	10	10 3	120	7 7	1	6 1	6	**
Sample input	Sample output											
5 2	10											
10 3	120											
7 7	1											
6 1	6											

61.	Write a program (WAP) that will find x^y (x to the power y) where x, y are positive integers. <table><tr><th>Sample input(x,y)</th><th>Sample output</th></tr><tr><td>5 2</td><td>25</td></tr><tr><td>2 0</td><td>1</td></tr><tr><td>6 1</td><td>6</td></tr><tr><td>0 5</td><td>0</td></tr></table>	Sample input(x,y)	Sample output	5 2	25	2 0	1	6 1	6	0 5	0	*		
Sample input(x,y)	Sample output													
5 2	25													
2 0	1													
6 1	6													
0 5	0													
62.	WAP that will find the GCD (greatest common divisor) and LCM (least common multiple) of two positive integers. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>5 7</td><td>GCD: 1 LCM: 35</td></tr><tr><td>12 12</td><td>GCD: 12 LCM: 12</td></tr><tr><td>12 32</td><td>GCD: 4 LCM: 96</td></tr></table>	Sample input	Sample output	5 7	GCD: 1 LCM: 35	12 12	GCD: 12 LCM: 12	12 32	GCD: 4 LCM: 96	**				
Sample input	Sample output													
5 7	GCD: 1 LCM: 35													
12 12	GCD: 12 LCM: 12													
12 32	GCD: 4 LCM: 96													
63.	WAP that will determine whether a number is prime or not. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>1</td><td>Not prime</td></tr><tr><td>2</td><td>Prime</td></tr><tr><td>11</td><td>Prime</td></tr><tr><td>39</td><td>Not prime</td></tr><tr><td>101</td><td>Prime</td></tr></table>	Sample input	Sample output	1	Not prime	2	Prime	11	Prime	39	Not prime	101	Prime	**
Sample input	Sample output													
1	Not prime													
2	Prime													
11	Prime													
39	Not prime													
101	Prime													
64.	WAP that will determine whether an integer is palindrome number or not. <table><tr><th>Sample input</th><th>Sample output</th></tr><tr><td>9</td><td>Yes</td></tr><tr><td>91</td><td>No</td></tr><tr><td>222</td><td>Yes</td></tr><tr><td>12321</td><td>Yes</td></tr><tr><td>110</td><td>No</td></tr></table>	Sample input	Sample output	9	Yes	91	No	222	Yes	12321	Yes	110	No	**
Sample input	Sample output													
9	Yes													
91	No													
222	Yes													
12321	Yes													
110	No													

65. WAP that will calculate following mathematical function for the input of x. Use only the series to solve the problem.

$$\text{Sinx} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \dots \dots \infty$$

<i>Sample input</i>	<i>Sample output</i>
1	0.841
2	0.909
3	0.141

66. Write a program that takes an integer number n as input and find out the sum of the following series up to n terms.

$$1 + 12 + 123 + 1234 + \dots$$

<i>Sample input</i>	<i>Sample output</i>
1	1
2	13
3	136
4	1370

**